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Shared genetic architecture and causality between autism spectrum disorder and irritable bowel syndrome, multisite pain, and fatigue

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Autism spectrum disorder (ASD) often co-occurs with functional somatic syndromes (FSS), such as irritable bowel syndrome (IBS), multisite pain, and fatigue. However, the underlying genetic mechanisms and causality have not been well studied. Using large-scale genome-wide association study (GWAS) data, we investigated the shared genetic architecture and causality between ASD and FSS. Specifically, we first estimated genetic correlations and then conducted a multi-trait analysis of GWAS (MTAG) to detect potential novel genetic variants for single traits. Afterwards, polygenic risk scores (PRS) of ASD were derived from GWAS and MTAG to examine the associations with phenotypes in the large Dutch Lifelines cohort. Finally, we performed Mendelian randomization (MR) to evaluate the causality. We observed positive genetic correlations between ASD and FSS (IBS: $r_g = 0.27$, adjusted $p = 2.04 \times 10^{-7}$; multisite pain: $r_g = 0.13$, adjusted $p = 1.10 \times 10^{-3}$; fatigue: $r_g = 0.33$, adjusted $p = 5.21 \times 10^{-9}$). Leveraging these genetic correlations, we identified 3 novel genome-wide significant independent loci for ASD by conducting MTAG, mapped to *NEDD4L*, *MFHAS1*, and *RP11-10A14.4*. PRS of ASD derived from both GWAS and MTAG were associated with ASD and FSS in Lifelines, and MTAG-derived PRS showed a bigger effect size, larger explained variance, and smaller p -values. We did not observe significant causality using MR. Our study found genetic associations between ASD and FSS, specifically with IBS, multisite pain, and fatigue. These findings suggest that a shared genetic architecture may partly explain the co-occurrence between ASD and FSS. Further research is needed to investigate the causality between ASD and FSS due to current limited statistical power of the GWASs.

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INTRODUCTION

Autism spectrum disorder (ASD) is a neurodevelopmental disorder affecting 1–1.5% of individuals [1, 2]. It is characterized by social-communication difficulties and restricted repetitive patterns of interests or behaviors. The symptoms of ASD usually appear in childhood and persist throughout life [3]. Genetic factors may explain part of the risk for ASD with a heritability of higher than 80% for ASD estimated by twin and family studies [4]. The most recent genome-wide association study (GWAS) with 18,381 cases and 27,969 controls has identified 3 susceptibility loci associated with clinically diagnosed ASD, explaining 2.45% of the variance in ASD [2].

Existing evidence indicates that many comorbid somatic problems in people with ASD are functional somatic symptoms (FSS), with irritable bowel syndrome (IBS), multisite chronic pain, and chronic fatigue syndrome among the most prominent [5, 6]. Similar to ASD, IBS, multisite chronic pain, and fatigue are also complex traits and genetic factors make substantial contributions to the etiology [7–11] with a heritability of up to 57% [12–14]. Recent GWASs have identified 89, 1748, and 1 single-nucleotide polymorphisms (SNPs) for IBS, multisite chronic pain and fatigue, respectively [15–17].

Epidemiological studies demonstrate associations between ASD and FSS [18]. Previously, we investigated comorbid somatic problems of ASD among 35,048 adults from the Netherlands. The results showed that adults with more ASD symptoms had more FSS problems including IBS, multisite pain, and fatigue [6]. As ASD and FSS are moderately to highly heritable, their comorbidity observed from epidemiological studies may suggest genetic overlap between ASD and FSS. For instance, a genetic correlation (r_g) of 0.33 between ASD and fatigue has been reported [2].

Two mechanisms may explain genetic correlations between ASD and FSS. One mechanism is horizontal pleiotropy, i.e., the genetic variants contribute directly to risk of both ASD and FSS or indirectly through another phenotype, indicating shared biological processes between ASD and FSS. The other mechanism is vertical pleiotropy, where there is a causal relationship between ASD and FSS [19].

Leveraging genetic correlations between ASD and FSS, analyses can be conducted that could improve statistical power to potentially detect new genetic variants for ASD and FSS and make inferences on causality with the advances in genetic methods using GWAS summary statistics. Current results from GWASs of ASD and FSS

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suggest potential limited statistical power in identifying significant genetic variants. Given the hypothesized genetic correlation between ASD and FSS, multi-trait analysis of GWAS (MTAG) leverages the genetic information of correlated traits. It can thus boost statistical power to detect novel genetic loci for each trait [20]. Also, a polygenic risk score (PRS) of a trait could be calculated in independent genotyped samples to explore its associations with genetically correlated phenotypes [21]. Besides, Mendelian randomization (MR) analyses are considered a promising way to explore potential causality of one phenotype on the other (and vice versa) using genetic variants as instrument variables (IV) [22].

Some of these methods have been applied previously to investigate the genetic overlap between ASD and FSS. For example, one study investigated causality between pain and neuropsychiatric disorders including ASD using MR, but did not find a causal relationship between pain and ASD [23]. Another study found that the PRS of ASD was associated with low pain tolerance, suggesting that a higher genetic risk of ASD might be associated with more severe pain symptoms [24]. Thus, while some evidence exists regarding the genetic correlation and causality between ASD and pain, the genetic relationships between ASD and IBS, as well as fatigue, have yet to be fully elucidated and whether these genetic correlations reflect a causal relationship remains largely unknown.

To address these gaps, we aimed to probe into the shared genetic etiology between ASD and FSS, specifically with respect to IBS, multisite pain, and fatigue. We first estimated the genetic correlations of ASD with IBS, multisite pain, and fatigue, and then conducted a MTAG leveraging the correlations to detect potentially novel genetic variants for each single trait. Afterwards, PRS of ASD derived from GWAS and MTAG were calculated to examine the associations with phenotypes of ASD and FSS in the large Dutch Lifelines cohort. Finally, we explored the causal relationships between ASD and FSS using MR. The results may help to unravel the shared genetic architecture and better understand shared biological pathways and causality underlying comorbid ASD and FSS conditions.

METHODS

Study design and data summary

The overall study design is shown in Fig. 1. We used summary statistics retrieved from publicly available GWASs, ASD from the Psychiatric

Genomics Consortium ($N_{\text{case}}/N_{\text{control}} = 18,381/27,969$) [2], IBS from the UK BioBank ($N_{\text{case}}/N_{\text{control}} = 53,400/433,201$) [15], multisite chronic pain from the UK BioBank ($N = 387,649$) [16], and fatigue from the UK BioBank ($N = 108,976$) [17]. Polygenic risk score analyses were conducted in the large Dutch Lifelines cohort.

Lifelines is a multi-disciplinary prospective population-based cohort study examining in a unique three-generation design the health and health-related behaviours of 167,729 persons living in the North of the Netherlands [25, 26]. It employs a broad range of investigative procedures in assessing the biomedical, socio-demographic, behavioural, physical and psychological factors which contribute to the health and disease of the general population, with a special focus on multi-morbidity and complex genetics. The participants were recruited between 2006 and 2013 at baseline and followed up from 2014 on-ward. The Lifelines protocol was approved by the University Medical Centre Groningen (UMCG) Medical ethical committee under number 2007/152 and all participants signed an informed consent form [27].

Phenotype measurements

Autism spectrum disorder problems. The Adult Social Behaviour Questionnaire (ASBQ) was used to measure the extent to which ASD problems were present in individuals [28, 29], which is a 44-item validated questionnaire to quantify both milder and more severe forms of ASD in adults. The ASBQ assesses six domains of autistic problems: reduced contact, reduced social insight, reduced empathy, violation of social conventions, difficulty with change, and stereotyped sensory-motor behaviors. For each item, a 3-point scale is used to indicate whether the behavior “does not describe you” (score 0), “infrequently describes you” (score 1), or “clearly applies to you” (score 2). A continuous total score was used in the analysis, with higher scores indicating that more ASD problems are present.

Irritable bowel syndrome. IBS was assessed by the Rome III IBS Diagnostic Questionnaire, which has high sensitivity and specificity for detecting IBS according to Rome III criteria [30–32]. The 27-item questionnaire measures the onset, frequency, and duration of bowel complaints (e.g., bowel movement, swollen belly) and other related symptoms (e.g., defecation). Participants were identified “having IBS” if they met the following four diagnostic criteria: (1) Having recurrent abdominal pain or discomfort at least 1 day per week; (2) With onset at least 6 months; (3) Having at least two of the following three additional symptoms: a. Improvement with defecation; b. Onset associated with a change in frequency of stool; c. Onset associated with a change in appearance or form of stool; (4) For women, abdominal pain or unpleasant feeling should not only occur during menstruation. Other participants were classified as “not having IBS”.

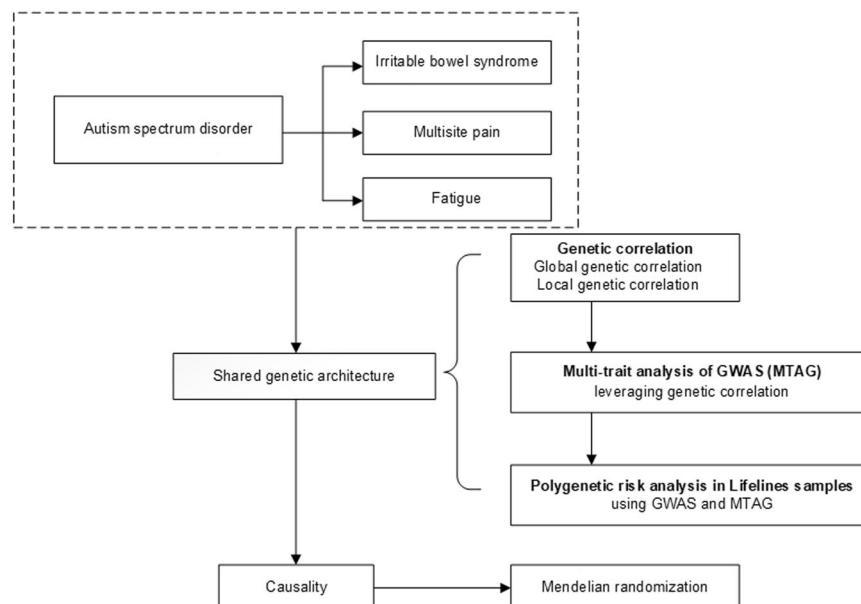


Fig. 1 Study design. GWAS genome-wide association studies.

Multisite pain. The Widespread Pain Index (WPI) was used to measure multisite pain. This 19-item checklist measures the number of painful body locations over the past 1 week (e.g., neck, hip, leg, etc.). Each affected location is scored 1 point, yielding a total score ranging from 0 to 19 [33, 34]. The item mean score ranging from 0 to 1 was calculated, with higher scores indicating greater multisite pain.

Fatigue. The Checklist Individual Strength survey was used to assess fatigue [35]. This 20-item questionnaire assesses four domains of fatigue-related symptoms in the past 2 weeks, including subjective fatigue, concentration, motivation, and physical activity [36]. Each item was rated on a 7-point Likert scale. Item mean score ranging from 1 to 7 was calculated, with higher scores indicating a higher level of fatigue.

Estimation of genetic correlations

We performed genome-wide genetic correlation analyses for ASD and FSS including IBS, multisite pain, and fatigue, respectively. First, global genetic correlations (r_g) were estimated from GWAS summary statistics by linkage disequilibrium score regression (LDSC) [37]. Global genetic correlation refers to the correlation of genetic effects between two related traits throughout the genome, highlighting the degree of pleiotropy and genetic overlap [37, 38]. We filtered genetic variants using the following criteria: imputation score (INFO) > 0.9 and minor allele frequency (MAF) > 0.01. Indels, strand-ambiguous SNPs, and SNPs with duplicated rs numbers were removed. We adjusted p -values for multiple testing by the false discovery rate (FDR) using the Benjamini-Hochberg procedure and a $p < 0.05$ was considered statistically significant.

The global genetic correlation describes the average effect of pleiotropy across all causal loci, but the underlying genetic architecture of correlations at individual loci can vary [19]. To gain better insight into genetic correlation at the locus level (within genome-wide partitioned LD blocks), local genetic correlations between ASD and FSS were calculated by the Local Analysis of coVariant Association (LAVA) tool. Unlike the global correlation analysis, the local approach enables us to identify and estimate genetic correlation in specific genomic regions, thereby providing insights into their local effects and shared genetic basis. The method has been detailed elsewhere [39]. Briefly, 2495 semi-independent LD blocks were defined using European-ancestry 1000 Genome data [39]. Then we conducted univariate association analyses on the local genetic signal of each trait. Regions that showed significant univariate associations for more than one trait were selected to test for local genetic correlation. For this purpose, we used a relatively lenient p -value threshold of 0.05. Finally, pairwise bivariate local genetic correlation analyses across the selected regions were estimated between ASD, IBS, multisite pain, and fatigue. For the correlation analyses, we adjusted p -values for multiple testing by FDR using the Benjamini-Hochberg procedure and a $p < 0.05$ was considered statistically significant.

Multi-trait analysis of GWAS

Because of the putative genetic correlations between ASD and FSS, the multi-trait analysis of GWAS (MTAG) was conducted to perform the meta-analysis of the four traits [20]. MTAG leverages the genetic correlations between multiple traits to generate trait-specific effect estimates for each SNP using GWAS summary statistics. As the authors describe, MTAG is robust to potential sample overlap between different GWASs, which is also an issue in our study, i.e., the three phenotypes of FSS were all from UK biobank. Current results from GWASs of ASD and FSS suggested potential statistical power issue to identify significant genetic variants. MTAG improves the effect estimates of each trait and boosts statistical power to detect genetic associations. Therefore, it can also improve the power of PRS [20]. We filtered genetic variants using default MTAG parameters [20]. Briefly, variants were restricted to those common to all four traits of the GWAS, with a MAF > 0.01. For each trait, we calculated the 90th percentile of the SNP sample-size distribution. All SNPs with a sample size below 75% of this calculated value were removed.

To identify genomic risk loci from the MTAG results, functional Mapping and Annotation (FUMA) analyses were conducted [40]. SNPs with p -value < 5×10^{-8} were considered genome-wide significant. Lead SNPs were LD-pruned based on a 250 kilobase (kb) window and $r^2 < 0.1$ using the 1000 Genome European population as a LD reference panel. Lead SNPs which were closer than 250 kb were merged into one genomic risk locus. ANNOVAR employed in FUMA was used to map SNPs to genes [41]. A locus that was not found to be significant in previous GWASs and was not in LD

with a previous locus in GWASs ($r^2 < 0.1$ within 1 megabase (Mb) window) for the phenotype was labeled as “novel”.

PRS analyses

To validate the shared genetic architecture between ASD and FSS, PRS analyses were conducted in an independent cohort the Lifelines Cohort. We included 14,134 participants with European ancestry from the Lifelines cohort with phenotypic and genetic data. DNA samples were genotyped using the Illumina Global Screening Array (N = 10,758) and Illumina CytoSNP12v2 array (N = 3376). After quality control, both genotyping datasets were then imputed at the Sanger imputation server using the Haplotype Reference Consortium panel r1.1 [42]. Details of genotyping, quality control, and imputation in Lifelines have been published elsewhere [43]. Two PRSs of ASD were calculated in the Lifelines samples based on the summary statistics of ASD GWAS and ASD MTAG results, respectively. PRS analyses were conducted to explore whether ASD-related genetic effects were associated with IBS, multisite pain, and fatigue based on the genetic profile, using independent European-based samples. PRSs based on both GWAS and MTAG of ASD were evaluated to further validate the shared genetics in external samples.

SNPs of low quality were removed from summary statistics (INFO ≤ 0.8 , MAF ≤ 0.01 , strand-ambiguous or duplicated SNPs). To obtain an independent set of SNPs, the LD-driven clumping procedure was performed in PLINK version 1.90 ($r^2 < 0.1$, kb = 250) using the 503 European samples from 1000 Genome projects as reference. Then weighted PRSs were calculated as the sum of risk allele dosages weighted by the corresponding effect size estimated in the GWAS results. PRSs were constructed across a range of p -value thresholds (5×10^{-8} , 5×10^{-7} , 5×10^{-6} , 5×10^{-5} , 5×10^{-4} , 5×10^{-3} , 0.05, 0.5, 1) and were then standardized using z-score transformations. Thereafter, we conducted a principal component analysis (PCA) on the resulting PRSs and used the first principal component as the final ASD PRS in subsequent association analyses [44]. This approach uses PCA to reweight the SNPs included in the PRS to achieve maximum variation across all the p -value thresholds. It avoids choosing one optimal p -value threshold, therefore controlling type-1 error and preventing overfitting.

Linear regression models were applied to examine the associations between ASD PRSs and phenotypes of ASD, IBS, multisite pain, and fatigue, adjusted for age, sex, arrays, and 10 genetic principal components that correct population structure. PRS and phenotype measures were standardized to Z-scores (mean = 0 and standard deviation = 1) to facilitate interpretation. We adjusted p -values for multiple testing by FDR using the Benjamini-Hochberg procedure and a $p < 0.05$ was considered statistically significant.

Bidirectional Mendelian randomization analyses

Bidirectional two-sample MR was conducted to infer putative causal relationships between ASD and FSS from GWAS summary statistics. MR estimates the causal effects of the exposure on the outcome using genetic variants as IVs [22]. We performed the method of random-effect inverse variance weighted (IVW) as the main analysis, and four complementary methods as sensitivity analyses, including MR Egger, weighted median, and median-based methods (simple and weighted) [45]. IVs were selected based on association statistics of the exposure GWAS summary statistics (Table S1). First, genome-wide significant SNPs ($p < 5 \times 10^{-8}$) were selected. For SNPs that were not available in outcome GWASs, a proxy SNP identified in high-LD ($r^2 > 0.7$) was used. If the proxies were not available (e.g., the only one significant SNP in the GWAS of fatigue is 1:64178756_C_T and it does not have an rs id or proxy), we chose SNPs that have a p -value less than 5×10^{-7} . Second, independent genetic variants were selected by LD clumping procedure at $r^2 < 0.001$ within 10 Mb. SNP alleles were harmonized between exposure and outcome summary statistics and palindromic SNPs with intermediate allele frequencies (0.42–0.58) were discarded. The Steiger filtering method was applied to remove potential invalid IVs that explain more variance in the outcome than the exposure [46]. To evaluate horizontal pleiotropy (i.e., the “exclusion restriction” assumption of MR [47]), heterogeneity tests by MR-Egger and IVW methods, and Egger-intercept tests were applied [48]. MR analyses were performed using the TwoSampleMR (version 0.5.6) package for R (version 4.0.5) [46, 49]. We adjusted p -values for multiple testing by FDR using the Benjamini-Hochberg procedure and a $p < 0.05$ was considered statistically significant.

Table 1. Global genetic correlations of ASD and FSS.

Phenotype 1	Phenotype 2	r_g	SE	p	Adjusted p^a
ASD	IBS	0.27	0.05	1.70×10^{-7}	2.04×10^{-7}
	Multisite pain	0.13	0.04	1.10×10^{-3}	1.10×10^{-3}
	Fatigue	0.33	0.06	3.47×10^{-9}	5.21×10^{-9}
IBS	Multisite pain	0.59	0.03	4.79×10^{-100}	2.87×10^{-99}
	Fatigue	0.52	0.04	6.53×10^{-31}	1.31×10^{-30}
Multisite pain	Fatigue	0.68	0.03	2.33×10^{-86}	3.99×10^{-86}

ASD autism spectrum disorder, FSS functional somatic syndrome, r_g genetic correlation, SE standard error, IBS irritable bowel syndrome.

^aAdjusted p : adjust p -values were calculated by false discovery rate using the Benjamini–Hochberg procedure.

Table 2. Statistical summary of GWAS and MTAG results.

	ASD		IBS		Multisite pain		Fatigue	
	GWAS	MTAG	GWAS	MTAG	GWAS	MTAG	GWAS	MTAG
Significant SNPs	93	127	89	793	1748	1908	1	1084
Lead SNPs	3	5	6	19	50	52	0	21
Genomic locus	3	5	6	19	46	48	0	21

GWAS genome-wide association studies, MTAG multi-trait analysis of GWAS, ASD autism spectrum disorder, IBS irritable bowel syndrome, SNP single nucleotide polymorphism.

RESULTS

Genetic correlation

We found significant genetic correlations between all the phenotype pairs. Regarding ASD, there was a positive genetic correlation with IBS ($r_g = 0.27$, adjusted $p = 2.04 \times 10^{-7}$), multisite pain ($r_g = 0.13$, adjusted $p = 1.10 \times 10^{-3}$), and fatigue ($r_g = 0.33$, adjusted $p = 5.21 \times 10^{-9}$) (Table 1). Additionally, we detected 21 positive local genetic correlations among the four phenotypes (Table S2). Regarding ASD, one was with multisite pain: chromosome 11: 75.45–76.52 Mb ($r_{g, local} = 0.61$, adjusted $p = 0.044$), and two were with IBS: (1) chromosome 13: 53.34–54.68 Mb ($r_{g, local} = 0.78$, adjusted $p = 0.019$) and (2) chromosome 16: 53.39–54.87 Mb ($r_{g, local} = 0.56$, adjusted $p = 0.036$).

Multi-trait analysis of GWAS

Given the significant genetic correlations between ASD and FSS, we improved the overall statistical power to detect novel loci by conducting MTAG analysis. A total of 6,506,975 overlapping SNPs were used in MTAG. After MTAG analysis, the number of lead SNPs in the GWAS result of ASD increased from 3 to 5, and the distribution of locations was more extensive (from chromosomes 8, 20 to chromosomes 5, 8, 18, 20) (Tables 2, 3, Fig. 2). Similarly, the number of lead SNPs for FSS phenotypes also increased and the distributions of locations were more extensive (Table 2, Fig. 2).

We identified 3 novel genome-wide significant independent loci for ASD by MTAG (Table 3). Among the 3 novel loci, 2 of them were mapped to protein-coding genes. In these 2 loci, the stronger signal was observed on chromosome 18 at the NEDD4L region (lead SNP rs63615960, $p = 1.81 \times 10^{-8}$). The other signal was observed on chromosome 8 mapped to the MFHAS1 and RP11-10A14.4 region (lead SNP rs12547493, $p = 3.32 \times 10^{-8}$).

Associations between PRSs of ASD and phenotypes

Characteristics of the total sample are shown in Table S3. Table 4 shows the results of the regression models examining the associations between GWAS and MTAG PRSs of ASD and phenotypes of ASD and FSS in Lifelines. We found that the effects of both ASD PRSs on phenotypes of ASD and FSS were positive. For PRS derived from ASD GWAS, each unit (equal to 1 SD) increase in PRS of ASD was associated with 1.12 times higher odds

of having IBS (for IBS: OR = 1.12, 95% CI: (1.03, 1.21)). Similarly, each unit (equal to 1 SD) increase in PRS of ASD was associated with 0.02 units increase in ASD, multisite pain, and fatigue (for ASD, $b = 0.023$, 95% CI: (0.006, 0.039); for multisite pain, $b = 0.020$, 95% CI: (0.002, 0.038); for fatigue, $b = 0.024$, 95% CI: (0.006, 0.042)). For PRS derived from MTAG result of ASD, the associations were more pronounced with larger b and explained variance, and smaller p -values, compared with PRS derived from ASD GWAS (Table 4).

Bidirectional MR

We did not observe significant causal relationships between ASD and FSS traits (Table 5). Sensitivity analyses using different MR methods yielded similar non-significant results. In forward MR, we extracted 3 IVs to examine the effect of ASD on FSS; in reverse MR, we extracted 4 IVs for IBS, 33 IVs for multisite pain, and 3 IVs for fatigue to examine the effects of FSS on ASD. The included IVs are presented in the supplementary materials (Table S1). The results of Egger-intercept tests and almost all heterogeneity analyses were not significant, suggesting a low risk of bias caused by horizontal pleiotropy (Tables S4, 5).

DISCUSSION

This study comprehensively investigated shared genetic architecture and causality between ASD and FSS using different genetic analyses and findings contribute to a better understanding of mechanisms underlying the co-occurrence between ASD and FSS. The results showed that ASD had positive genetic correlations with IBS, multisite pain, and fatigue. Leveraging these genetic correlations, we identified previously unknown risk loci for ASD by conducting MTAG, and we further validated genetic associations between ASD and FSS in an independent European sample. We did not find a significant causal relationship between ASD and FSS by MR.

Converging with previous evidence supporting the phenotypic co-occurrence of ASD and FSS [18, 23, 24, 50], we found positive genetic correlations between them. Moreover, we detected 3 positive genetic correlations at locus level, which gives a closer insight into underlying shared genetic architecture between ASD and FSS. Based on these genetic correlations, we identified novel genetic variants and mapped

Table 3. Genome-wide significant loci associated with ASD.

Genomic locus	rsID	Chromosome	Position	P value in GWAS	P value in MTAG	Start of position	End of position	Mapped genes
GWAS								
1	rs910805	20	21248116	2.04×10^{-9}	2.21×10^{-9}	20118473	22105020	XRN2, NKX2-4
2	rs10099100 ^a	8	10576775	1.07×10^{-8}	NA ^a	10571591	10583506	RP1L1, SOX7
3	rs71190156 ^b	20	14836243	2.75×10^{-8}	NA ^b	14697386	14836243	MACROD2,
MTAG								
1	rs910805 ^c	20	21248116	2.04×10^{-9}	2.21×10^{-9}	21117840	21474473	XRN2, NKX2-4
2	rs60410697 ^d	8	10572617	6.91×10^{-8}	2.73×10^{-8}	10566591	10802146	RP1L1, C8orf74, SOX7
3	rs416223	5	103991476	3.84×10^{-7}	2.44×10^{-8}	103791044	104082179	–
4	rs63615960	18	55881401	1.60×10^{-6}	1.81×10^{-8}	55855025	55898199	NEDD4L
5	rs12547493	8	8661534	4.68×10^{-6}	3.32×10^{-8}	8310508	9033744	MFHAS1, RP11-10A14.4

ASD autism spectrum disorder, GWAS genome-wide association studies, MTAG multi-trait analysis of GWAS.

^aNot available. The lead SNP in GWAS was removed from the MTAG analysis due to strand ambiguity.

^bNot available. The lead SNP in GWAS was removed from the MTAG analysis because it did not overlap among the four phenotypes.

^cThe same lead SNP as found in GWAS.

^dIn linkage disequilibrium with SNP rs10099100 found in GWAS ($r^2 = 0.83$).

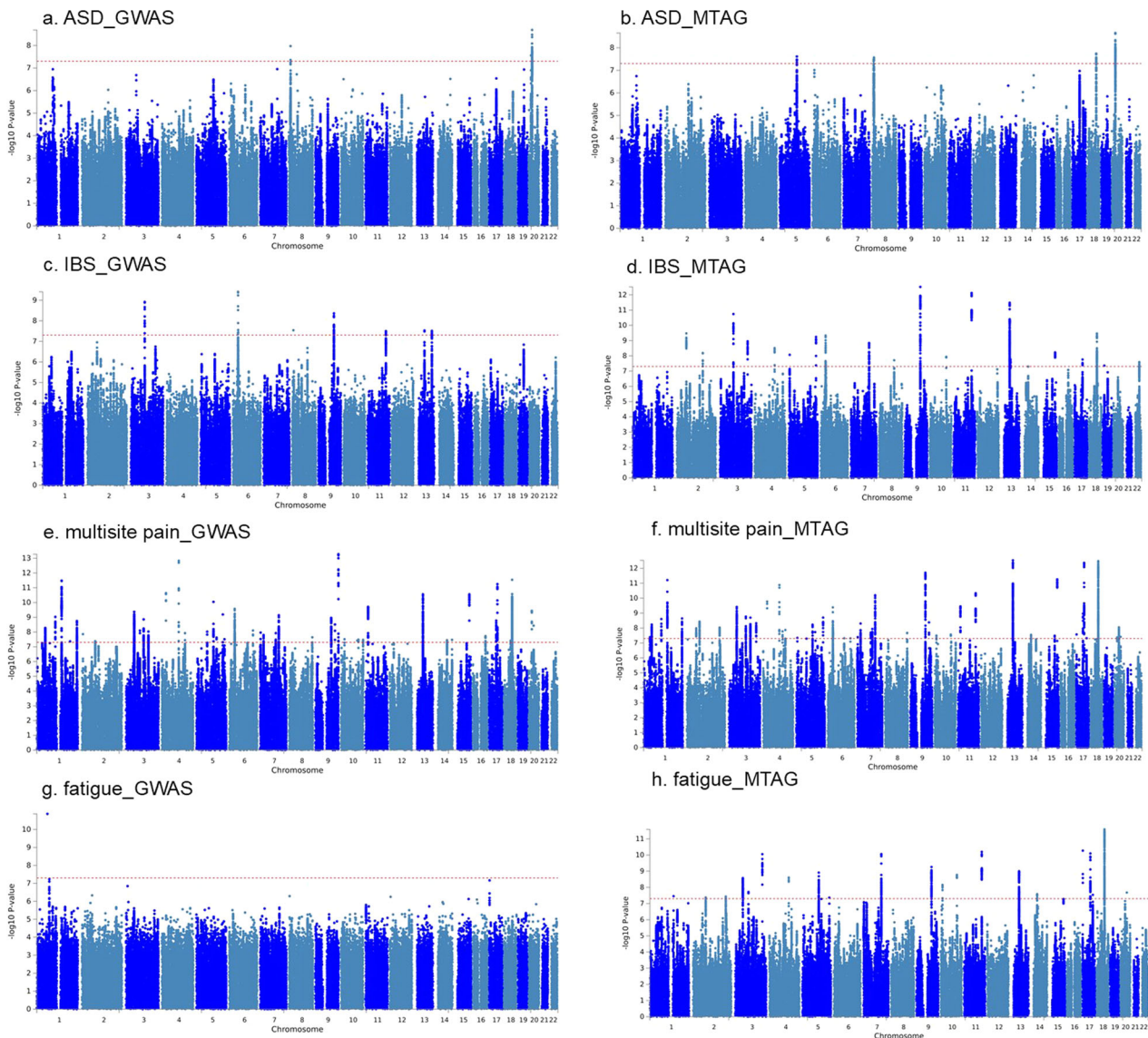


Fig. 2 Manhattan plots of GWAS and MTAG results. The panels **a**, **c**, **e**, and **g** display the results of GWAS and panels **b**, **d**, **f**, and **h** for MTAG. The X-axis is the chromosomal position, and the Y-axis is negative $\log_{10}$ transformed p -values for each single nucleotide polymorphism (SNP). The red line indicates genome-wide significance ($p = 5 \times 10^{-8}$). GWAS genome-wide association studies, MTAG multi-trait analysis of GWAS, ASD autism spectrum disorder, IBS irritable bowel syndrome.

Table 4. PRS effect on phenotypes.

Outcomes	ASD PRS calculated from GWAS results				ASD PRS calculated from MTAG results			
	Effect size [95% CI]	p	Adjusted p ^a	Adjusted R ² (%)	Effect size [95% CI]	p	Adjusted p ^a	Adjusted R ² (%)
ASD	0.023	0.007**	0.009**	1.9	0.041	<0.001***	<0.001***	2.0
(b [95% CI])	[0.006, 0.039]				[0.025, 0.058]			
IBS	1.12	0.006**	0.009**	3.0 ^b	1.20	<0.001***	<0.001***	3.2 ^b
(OR [95% CI])	[1.03, 1.21]				[1.11, 1.31]			
Multisite pain	0.020	0.032*	0.032*	2.6	0.053	<0.001***	<0.001***	2.8
(b [95% CI])	[0.002, 0.038]				[0.035, 0.071]			
Fatigue	0.024	0.008**	0.009**	2.5	0.059	<0.001***	<0.001***	2.8
(b [95% CI])	[0.006, 0.042]				[0.041, 0.077]			

PRS polygenic risk score, ASD autism spectrum disorder, GWAS genome-wide association studies, MTAG multi-trait analysis of GWAS, IBS irritable bowel syndrome, CI confidence interval, OR odds ratio.

*p < 0.05, **p < 0.01, ***p < 0.001.

^aAdjusted p: adjust p-values were calculated by false discovery rate using the Benjamini-Hochberg procedure.

^bMcFadden's R squared.

Table 5. The bidirectional MR analysis of ASD and FSS phenotypes.

Exposure	Outcome	MR method	Forward				Reverse			
			Number of IVs	beta	SE	p	Adjusted p ^a	Number of IVs	beta	SE
ASD	IBS	IVW	3	0.066	0.05	0.194	0.470	4	0.279	0.17
		MR Egger	3	-0.245	0.45	0.681	0.817	4	-0.006	2.97
		Weighted median	3	0.078	0.06	0.205	0.470	4	0.376	0.20
		Simple mode	3	0.103	0.08	0.336	0.504	4	0.418	0.30
Multisite pain		Weighted mode	3	0.104	0.09	0.356	0.509	4	0.421	0.28
		IVW	3	0.002	0.03	0.954	0.987	33	0.253	0.19
		MR Egger	3	0.372	0.15	0.245	0.470	33	0.156	0.97
		Weighted median	3	-0.008	0.03	0.773	0.882	33	0.390	0.24
Fatigue		Simple mode	3	-0.031	0.04	0.479	0.653	33	0.609	0.54
		Weighted mode	3	-0.025	0.04	0.560	0.700	33	0.539	0.53
		IVW	3	0.017	0.07	0.794	0.882	3	0.816	0.42
		MR Egger	3	0.746	0.27	0.218	0.470	3	-1.581	1.69
		Weighted median	3	0.063	0.04	0.129	0.470	3	1.023	0.51
		Simple mode	3	0.074	0.05	0.282	0.470	3	1.146	0.64
		Weighted mode	3	0.074	0.05	0.254	0.470	3	1.146	0.63
										0.209

MR Mendelian randomization, ASD autism spectrum disorder, FSS functional somatic syndrome, IV instrumental variable, SE standard error, IBS irritable bowel syndrome, IVW inverse variance weighted.

^aAdjusted p: adjust p-values were calculated by false discovery rate using the Benjamini-Hochberg procedure.

genes of ASD using MTAG, pointing to shared genetic mechanisms of co-occurrence. We highlight the potentially interesting functions of MFHAS1 and NEDD4L. MFHAS1 is known to be involved in the regulation of inflammation [51, 52]. Neuroinflammation and altered inflammatory responses have been established as key mechanisms in the development and maintenance of ASD [53, 54]. Gut inflammation is also considered a key factor of comorbid gastrointestinal disorders in ASD [18]. Additionally, an animal study suggested that MFHAS1 was related to cognitive impairment, which is strongly linked to ASD [55, 56]. Another novel gene is NEDD4L, which is known as an epilepsy-associated gene by mediating neuronal circuit activity [57]. It was also identified from single-cell sequencing of autism cortical tissue, suggesting it may be modulated as a downstream target in ASD [58, 59]. Besides, NEDD4L is associated with sensitivity to inflammatory pain by regulating nociceptive sensations [60, 61]. Dysfunction of NEDD4L can cause the occurrence of neuropathic pain [62]. Our findings show that the MTAG analysis improves the power to identify previously unknown risk genes involved in the shared genetic basis between ASD and FSS, and advances our understanding of the co-occurrence mechanisms.

Our results showed that PRSs of ASD were associated with the phenotypes of ASD and FSS. One previous study reported the association between PRS of ASD and ASD phenotype at age 6 years [7, 8]. The proportion of explained variation was 2.4%, which was comparable with our results (adjusted $R^2 = 1.9\%$). We added to the literature that PRS of ASD is also associated with presence of IBS, and severity of multisite pain and fatigue. Consistent with our results, Louis et al. found that PRS of ASD was associated with sensory processing issues [24]. Their findings showed that higher polygenic loading of ASD was related to lower pain tolerance, thus possibly leading to more severe pain symptoms [24]. Our findings further validated genetic associations between ASD and FSS in external samples.

Our study did not find any evidence supporting a causal relationship between ASD and FSS. This is in line with an earlier study that did not find a causal relationship between pain and ASD using the MR method and the same GWASs [23]. To the best of our knowledge, causality of ASD in relation to IBS and fatigue have not been explored using GWAS data in the literature. A potential reason for the lack of evidence for causality in our study could be the limited statistical power of the GWASs of particularly ASD and fatigue, given insufficient independent significant SNPs in the GWAS (7 for ASD and 0 for fatigue). To solve the problem, we used a relatively lenient p -value threshold of 5×10^{-7} for the IV of ASD and fatigue. However, it still resulted in limited IVs (3 for ASD and 3 for fatigue). Clearly, additional, more strongly powered GWASs of ASD and FSS with more IVs are warranted for future studies to examine the causality using GWAS.

Our results should be interpreted in light of several limitations. First, although the novel genetic variants identified by MTAG revealed some potential mechanisms of co-occurrence of ASD and FSS, they still need to be replicated in future studies to avoid false positives. Second, although we used GWAS summary statistics with the largest sample sizes up to date, the sample sizes were still relatively small, which limits the power of PRS and MR. Third, our study focused on individuals of European ancestry. The genetic relationship between ASD and FSS in other populations remain to be studied. Another limitation is that the phenotypes used in PRS analyses in our study differed from the phenotypes in the original GWASs. ASD was measured based on a continuous ASD symptom scale, whereas the GWAS used clinically diagnosed ASD [2]. Multisite pain in our study was assessed over the past week, whereas the GWAS evaluated it over a three-month period [16]. Fatigue was measured over the last two weeks in both our study and the GWAS, but was a single item worded "tired or little energy" in the original GWAS [17] while currently we used a 20-item scale measuring fatigue-related symptoms. Our findings thus warrant replication with

measures of diagnosed ASD, chronic multisite pain, and chronic fatigue consistent with the original GWASs or await additional GWASs better aligned with our current phenotypes.

CONCLUSION

Our study found genetic associations between ASD and FSS, specifically with IBS, multisite pain, and fatigue. These findings suggest that a shared genetic architecture may partly explain the co-occurrence between ASD and FSS. Further research is needed to investigate the causality between ASD and FSS due to current limited statistical power of the GWASs.

DATA AVAILABILITY

Data may be obtained from a third party and are not publicly available. Researchers can apply to use the Lifelines data used in this study. More information about how to request Lifelines data and the conditions of use can be found on their website (<https://www.lifelines-biobank.com/researchers/working-with-us>).

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AUTHOR CONTRIBUTIONS

YL and TX contributed equally. YL performed the data analysis and wrote the manuscript; TX conceptualized and supervised the project and reviewed the manuscript; MV reviewed the manuscript; HS supervised the project and reviewed the manuscript; CAH supervised the project and reviewed the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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